

Structural healing on quantitative ultrasound.

eCTM ligaments rebuilt tissue architecture over 12 weeks; standard-of-care ligaments deteriorated.

Background

OBJECTIVE

Return to sport depends on structural healing, not just symptom relief. Standard B-mode ultrasound reads out qualitatively, with no reliable early signal of repair. We asked whether a quantitative, texture-based Tissue Remodeling Score (TRS) could detect objective architectural recovery after a single injection of cryopreserved umbilical allograft (equicenta® CTM) vs. standard of care.

Methods

IMAGING & SCORING

B-mode cine loops (Butterfly iQ3) at Weeks 0, 2, 4 and 12. Six texture features combined into a composite TRS, normalized to 50 at baseline. Thresholds: Responding (55+) · Equivocal (45–55) · Not Responding (<45). Segmentation was blinded to treatment.

TWO-LAYER MODEL OF HEALING

TRS decomposes into MS Core (deep scaffold: central dark area, dark intensity) and Quality Layer (fiber alignment, echo intensity). Responders showed early MS Core recovery by Week 4, then later Quality Layer remodeling — deep architecture reorganizing before surface tissue matures.

SAMPLE & VALIDATION

46 QC-passed injuries: 22 eCTM vs. 11 SOC ligaments plus 13 exploratory tendons. Segmentation reliability (25 ROIs / 8 cases): Dice 0.931. Tendon TRS correlated with lameness at Weeks 2–4 ($r=-0.74$, $p<0.02$). In an exploratory analysis, ligament TRS trended toward an association with 26-week performance (Cohen's $d=0.96$, $p=0.073$ — not significant).

Interpretation

TISSUE-LEVEL EVIDENCE OF HEALING

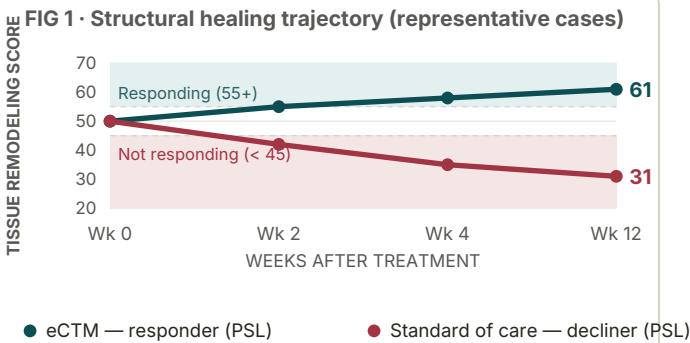
eCTM ligaments built responding TRS scores (55+) across Weeks 2–12; SOC ligaments drifted below Not-Responding (<45). The healing signature was sequential — MS Core first, Quality Layer following — consistent with a biodegradable allograft releasing reparative proteins over time. In an exploratory, hypothesis-generating analysis, ligament TRS trended toward an association with 26-week performance (Cohen's $d=0.96$, $p=0.073$ — not statistically significant): a signal that structural healing may precede return to work, not a confirmed predictor.

Results

LIGAMENTS REACHING THE RESPONDING TRS THRESHOLD BY WEEK 12

36% : 0%

8 / 22 eCTM vs. 0 / 11 standard of care
 $p=0.013$ · Cohen's $d=0.98$



TRS over 12 weeks in matched proximal suspensory ligament (PSL) lesions. The eCTM case rose from baseline (50) into the Responding zone by Week 2 and reached 61; the standard-of-care case dropped below the Not-Responding threshold by Week 4 and reached 31. Illustrative trajectories from Supplementary Figure S1.

Healing you can measure, not just observe.

Equine Performance Labs

A blinded quantitative-ultrasound biomarker showed eCTM ligaments rebuilding tissue architecture over 12 weeks — a structural signature standard of care did not produce.